

VAMSI KRISHNA KOLLIPARA

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SUMMARY

AI Full-Stack Engineer with 2+ years shipping production React/TypeScript frontends and Node.js/Express backends on PostgreSQL, Redis, and Azure. Built an enterprise CRM with calendar, email (Microsoft Graph), and Claude-powered conversation analysis serving 8 roles daily, plus a RAG knowledge assistant on Pinecone vector search. Strong on enterprise auth (OAuth 2.0, JWT, Azure AD SSO, Row-Level Security) and secure API design. MS in Computer Science. 3 published PyPI packages.

SKILLS

Languages: JavaScript, TypeScript, Python, SQL, PowerShell, HTML, CSS

Frontend: React, Next.js, Vite, Tailwind CSS, shadcn/ui, D3.js, Mapbox GL, Responsive UI

Backend: Node.js, Express.js, REST APIs, GraphQL, Webhooks, RabbitMQ, Edge Functions

Databases: PostgreSQL, SQL Server, Oracle, Redis, Pinecone, Supabase

AI & LLM: Anthropic Claude API, OpenAI API, LangChain, RAG, Vector Search, Embeddings, Prompt Engineering

Auth & Security: OAuth 2.0, JWT, Session Management, Azure AD SSO, Row-Level Security, RBAC, Encryption at Rest/Transit

Cloud & DevOps: Azure, AWS, Docker, GitHub Actions, CI/CD, Vercel

Tools: Git, Postman, Jest, React Testing Library, Pytest

EXPERIENCE

AI Full-Stack Engineer, AIDM — Software Consultancy | Remote Jul 2025 – Present

- Architected and shipped an enterprise operations platform (React 18, TypeScript, Vite, Tailwind, shadcn/ui, GraphQL) on Node.js/Express services and Supabase Postgres, deployed via GitHub Actions across AWS and Vercel — consolidating CRM, applicant tracking, email campaigns, training/LMS, and analytics into one SPA used daily across 8 organizational roles for 10+ months.
- Designed LLM-powered NLU services on Anthropic Claude (via Supabase Edge Functions) for client-relationship workflows: automated PDF/DOCX résumé screening with fit scoring and reasoning, one-click CRM-account and applicant summaries, action-item extraction, and natural-language insights across a 24-chart analytics system, with DB-level result caching and per-user rate limiting.
- Built an audio-to-notes pipeline using OpenAI Whisper for transcription of uploaded recordings and in-browser live audio capture, with downstream Claude processing for summarization and action-item extraction — output stored both as CRM notes attached to accounts/contacts and as a searchable transcript table.
- Implemented a CRM with stage-tracked opportunity pipeline, account/contact management, territory assignment, and a FullCalendar visit planner with Leaflet + OSRM route optimization — covering the full advisor-style workflow from prospect to follow-up.
- Built an email integration and campaign engine on Microsoft Graph API for organizational sending, with dynamic CRM-driven templates, business-day scheduling, transactional delayed queues, and webhook-driven inbound reply tracking.
- Hardened the platform with enterprise auth and security: OAuth 2.0 + Azure AD SSO with identity linking, JWT-verified Edge Function endpoints, session management, and a centralized 8-role RBAC matrix enforced at the database layer via Postgres Row-Level Security.
- Integrated semantic search using vector embeddings to surface relevant CRM history, prior meeting context, and policy documents inline in the user workflow.

Graduate Research & Teaching Assistant, Indiana State University | Terre Haute, IN Aug 2023 – May 2025

- Built a RAG-powered knowledge assistant (LangChain, OpenAI API, Pinecone vector DB) for university staff to query academic policies and HR procedures in natural language, with configurable semantic retrieval, chunk-overlap tuning, and vector indexing — production patterns directly applicable to advisor-facing client/policy NLU.
- Built internal web applications (React, Node.js, Express, PostgreSQL) including a course catalog portal, faculty directory, and student resource dashboard used by academic departments, with JWT session auth and REST APIs.
- Open-sourced WebWeaver, a Python crawling library, and used it to ingest 40,000+ university URLs into a curated knowledge base of ~8,000 documents powering the RAG assistant's semantic search.

Full-Stack Developer, Cognizant Technology Solutions | Hyderabad, India

Jan 2022 – Jul 2023

- Developed React-based portal modules for an enterprise energy management platform, delivering real-time usage analytics, billing history, and outage status with interactive D3.js visualizations to residential and commercial customers.
- Built Node.js/Express REST APIs integrated with Oracle and SAP backends to surface smart meter telemetry data, achieving sub-200ms p95 latency through Redis caching and connection pooling.
- Built an internal outage reporting dashboard (React, PostgreSQL, Mapbox GL) aggregating grid sensor data to help operations teams identify fault zones and coordinate field crew dispatch.
- Optimized SQL queries on multi-million-row smart meter and billing tables, reducing average execution time by 30% through composite indexing, join restructuring, and eliminating N+1 patterns.
- Contributed to integrating grid event feeds into the customer notification system, building Node.js/RabbitMQ services that triggered real-time outage alerts via email and SMS.
- Wrote unit and integration tests (Jest, React Testing Library) achieving 85%+ code coverage across frontend and API layers.

EDUCATION

Master of Science in Computer Science

Indiana State University, Terre Haute, IN | GPA: 3.66

Aug 2023 – May 2025

Bachelor of Technology in Electronics & Communication Engineering

Vel Tech University, Chennai, India | GPA: 9.01/10

Jun 2018 – May 2022

PROJECTS & OPEN-SOURCE

GitBar — macOS Menubar Git Dashboard (PyPI)

Mar 2026

- Built and published an open-source macOS menubar app providing a unified Git dashboard aggregating PRs, issues, CI/CD status, and local repo health across GitHub, GitLab, and Bitbucket. Tech: Python, PyObjC, GitHub/GitLab REST APIs.

GadgetBox — Cross-Platform Developer Utilities (PyPI)

Feb 2026

- Published a cross-platform system tray app bundling 12 developer utilities (JSON formatter, JWT decoder, UUID generator, Base64, hash, regex tester, etc.) with intelligent clipboard auto-detection. Tech: Python, pystray, tkinter.

WebWeaver — Web Scraping & Crawling Library (PyPI)

Sep 2024

- Open-source Python library for configurable web crawling with URL validation, deduplication, robots.txt compliance, and recursive/breadth-first modes; powers the data ingestion layer behind the ISU RAG chatbot. Tech: Python, asyncio, aiohttp.